

Unit 1 Linear Maps

Notation F, V, W

- $\mathbb{R}$ denotes the set of all real numbers.
- $\mathbb{C}$ denotes the set of all complex numbers.
- F denotes $\mathbb{R}$ or $\mathbb{C}$.
- $\mathbb{R}^n$ and $\mathbb{C}^n$ are vector spaces of n -tuples.
- V and W denote vector spaces over F .
- The set of all linear maps from V to W is denoted $\mathcal{L}(V, W)$.
- The set of all polynomials with coefficients in F is $P(F)$.

The Vector Space of Linear Maps

Definition (1) p.2

A linear map from V to W is a function $T: V \rightarrow W$ with the following properties:

Additivity: $T(u+v) = Tu + Tv$
for all $u, v \in V$;

Homogeneity: $T(\lambda v) = \lambda(Tv)$ for all $\lambda \in F$
and all $v \in V$.

Eg 1.2 p.4

Let V be a finite-dimensional vector space over F , and $\{v_1, \dots, v_n\}$

be a basis of V . Define a map $T: V \rightarrow F^n$ by $Tv = (x_1, \dots, x_n)$,
for every $v \in V$, with $x_i \in F$. Show
that T is a linear map.

soln:

Let $u, w \in V$.

Then there exist scalars $\lambda_i \in F$
and $\mu_i \in F$, ($i=1, \dots, n$) such

$$u = \lambda_1 v_1 + \dots + \lambda_n v_n$$

$$w = \mu_1 v_1 + \dots + \mu_n v_n,$$

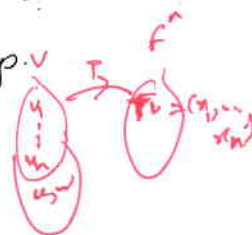
because $\{v_1, \dots, v_n\}$ is a basis
of V .

$$\begin{aligned} T(u+w) &= T(\lambda_1 v_1 + \dots + \lambda_n v_n + \mu_1 v_1 + \dots + \mu_n v_n) \\ &= T((\lambda_1 + \mu_1) v_1 + \dots + (\lambda_n + \mu_n) v_n) \\ &= (\lambda_1 + \mu_1, \dots, \lambda_n + \mu_n) \\ &= (\lambda_1, \dots, \lambda_n) + (\mu_1, \dots, \mu_n) \\ &= Tu + Tw \end{aligned}$$

If $\lambda \in F$, then

$$\begin{aligned} T(\lambda u) &= T(\lambda(\lambda_1 v_1 + \dots + \lambda_n v_n)) \\ &= T(\lambda \lambda_1 v_1 + \dots + \lambda \lambda_n v_n) \\ &= (\lambda \lambda_1, \dots, \lambda \lambda_n) \\ &= \lambda(\lambda_1, \dots, \lambda_n) \\ &= \lambda(Tu) \end{aligned}$$

Therefore, T is a linear map.



Eg 1.3 p.5

Suppose that $v = (\lambda_1, \dots, \lambda_n) \in F^n$ and $u = (\mu_1, \dots, \mu_n) \in F^n$. Then

the product $v \cdot u$ is defined by

$v \cdot u = (\lambda_1 \mu_1, \dots, \lambda_n \mu_n)$. Suppose

$T: R^3 \rightarrow R$ is defined by $Tv = v \cdot x$

where $x = (1, 2, -1)$. Show that

T is linear.

Solu:

Let $v, u \in R^3$, $v = (\lambda_1, \lambda_2, \lambda_3)$ and

$u = (\mu_1, \mu_2, \mu_3)$. Then

$$T(u+v) = (u+v) \cdot x$$

$$T(u+v) = (\lambda_1 + \mu_1, \lambda_2 + \mu_2, \lambda_3 + \mu_3) \cdot (1, 2, -1)$$

$$= (\lambda_1 + \mu_1)(1) + (\lambda_2 + \mu_2)(2) + (\lambda_3 + \mu_3)(-1)$$

$$= (\lambda_1)(1) + (\mu_1)(1) + (\lambda_2)(2) + (\mu_2)(2)$$

$$+ (\lambda_3)(-1) + (\mu_3)(-1)$$

$$= ((\lambda_1)(1) + (\lambda_2)(2) + (\lambda_3)(-1))$$

$$+ ((\mu_1)(1) + (\mu_2)(2) + (\mu_3)(-1))$$

$$= v \cdot x + u \cdot x$$

$$= Tv + Tu$$

If $\lambda \in F$, then

$$T(\lambda v) = (\lambda v) \cdot x$$

$$= (\lambda \lambda_1, \lambda \lambda_2, \lambda \lambda_3) \cdot (1, 2, -1)$$

$$= (\lambda \lambda_1)(1) + (\lambda \lambda_2)(2) + (\lambda \lambda_3)(-1)$$

$$= \lambda (\lambda_1 + 2\lambda_2 + (-1)\lambda_3)$$

$$= \lambda (v \cdot x)$$

$$= \lambda (Tv)$$

$\therefore T$ is linear.



S.A 1.1 No. 1 p.8 AX

Suppose $b, c \in R$. Define $T: R^3 \rightarrow R^2$

by $T(x, y, z) = (2x - 4y + 3z + b, 6x + cxy z)$

Show that T is linear if and only if

$b = c = 0$.

Solu:

Suppose that T is linear.

$$T((x_1, y_1, z_1) + (x_2, y_2, z_2))$$

$$= T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$$

$$\text{where } (x_1, y_1, z_1) + (x_2, y_2, z_2)$$

$$= (x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$T((x_1, y_1, z_1) + (x_2, y_2, z_2))$$

$$= T(x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$= (2(x_1 + x_2) - 4(y_1 + y_2) + 3(z_1 + z_2) + b,$$

$$6(x_1 + x_2) + c(x_1 + x_2)(y_1 + y_2)(z_1 + z_2))$$

Next,

$$T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$$

$$= (2x_1 - 4y_1 + 3z_1 + b, 6x_1 + c x_1 y_1 z_1)$$

$$+ (2x_2 - 4y_2 + 3z_2 + b, 6x_2 + c x_2 y_2 z_2)$$

$$= (2(x_1 + x_2) - 4(y_1 + y_2) + 3(z_1 + z_2) + 2b,$$

$$6(x_1 + x_2) + c(x_1 y_1 z_1 + x_2 y_2 z_2))$$

We obtain, $b = 2b$

$$c(x_1 + x_2)(y_1 + y_2)(z_1 + z_2) = c(x_1 y_1 z_1 + x_2 y_2 z_2)$$

Thus b and c must be zero to satisfy these equations.

Conversely, suppose that $b = c = 0$

$$T(x, y, z) = (2x - 4y + 3z, 6x)$$

Let $(x_1, y_1, z_1), (x_2, y_2, z_2) \in R^3$,

$$(x_1 + x_2, y_1 + y_2, z_1 + z_2) \in R^3$$

$$T(x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$= (2(x_1 + x_2) - 4(y_1 + y_2) + 3(z_1 + z_2), 6(x_1 + x_2))$$

$$= (2x_1 + 2x_2 - 4y_1 - 4y_2 + 3z_1 + 3z_2, 6x_1 + 6x_2)$$

$$= (2x_1 - 4y_1 + 3z_1, 6x_1) + (2x_2 - 4y_2 + 3z_2, 6x_2)$$

$$= T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$$

Let $(x, y, z) \in \mathbb{R}^3$ and $c \in \mathbb{R}$

$$c(x, y, z) = (cx, cy, cz)$$

$$T(c(x, y, z)) = T(cx, cy, cz)$$

$$= (2(cx) - 4(cy) + 3(cz), 6(cx))$$

$$= c(2x - 4y + 3z, 6x)$$

$$= cT(x, y, z)$$

$\therefore T$ is linear.

S.A 1.1 No-3 p-8 ASS

Suppose $T \in \mathcal{L}(F^n, F^m)$. Show that

there exist scalar $A_{j,k} \in F$ for

$j = 1, \dots, m$ and $k = 1, \dots, n$ such

$$\text{that } T(x_1, \dots, x_n) = (A_{1,1}x_1 + \dots + A_{1,n}x_n, \dots, A_{m,1}x_1 + \dots + A_{m,n}x_n).$$

for every $(x_1, \dots, x_n) \in F^n$.

Soln:

Let the standard basis of F^n be $\{e_1, e_2, \dots, e_n\}$.

$$e_1 = (1, 0, \dots, 0), e_2 = (0, 1, \dots, 0), \dots,$$

$$e_n = (0, 0, \dots, 1)$$

For any $x \in F^n$, $x = (x_1, \dots, x_n)$

$$x = x_1 e_1 + \dots + x_n e_n$$

$$T(x) = T(x_1 e_1 + \dots + x_n e_n)$$

$$= x_1 T(e_1) + \dots + x_n T(e_n)$$

Let the standard basis of F^m be

$$\{f_1, f_2, \dots, f_m\}. f_1 = (1, 0, \dots, 0), f_2 = (0, 1, \dots, 0), \dots, f_m = (0, 0, \dots, 1)$$

$$T(e_1) = A_{1,1}f_1 + A_{2,1}f_2 + \dots + A_{m,1}f_m \quad (2)$$

$$T(e_2) = A_{1,2}f_1 + A_{2,2}f_2 + \dots + A_{m,2}f_m$$

$\vdots$

$$T(e_n) = A_{1,n}f_1 + A_{2,n}f_2 + \dots + A_{m,n}f_m$$

where $A_{j,k}$ are coefficient scalars
 $j = 1, \dots, m, k = 1, \dots, n$.

$$T(x) = x_1(A_{1,1}f_1 + \dots + A_{m,1}f_m) + x_2(A_{1,2}f_1 + \dots + A_{m,2}f_m) + \dots + x_n(A_{1,n}f_1 + \dots + A_{m,n}f_m)$$

$$T(x) = (A_{1,1}x_1 + A_{1,2}x_2 + \dots + A_{1,n}x_n)f_1 + \dots + (A_{m,1}x_1 + A_{m,2}x_2 + \dots + A_{m,n}x_n)f_m$$

$$T(x) = (A_{1,1}x_1 + \dots + A_{1,n}x_n, \dots, A_{m,1}x_1 + \dots + A_{m,n}x_n).$$

ASS
S.A. 1.1 No-5 p.6 ASS

Prove that $\mathcal{L}(V, W)$ is a vector space.

Soln:

Let $S, T, R \in \mathcal{L}(V, W)$, $a, b \in F$.

For any $v \in V$.

$$(a) (S+T)(v) = S(v) + T(v) \quad (\because T \text{ is linear})$$

$$= T(v) + S(v)$$

$$= (T+S)(v)$$

$$\therefore S+T = T+S$$

$$(b) ((S+T)+R)(v)$$

$$= (S+T)(v) + R(v)$$

$$= [S(v) + T(v)] + R(v)$$

$$= S(v) + [T(v) + R(v)]$$

$$= S(v) + (T+R)(v)$$

$$= [S+(T+R)](v)$$

$$\therefore (S+T)+R = S+(T+R).$$

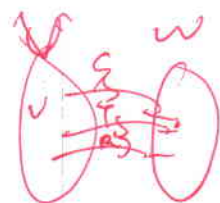
(c) Zero linear map $O \in \mathcal{L}(V, W)$.

$$O(v) = 0_W$$

$$(T+O)(v) = T(v) + O(v) = T(v) + 0 = T(v)$$

$$\therefore T+O = T$$

$\therefore O$ is additive identity.



(d) For $T \in L(V, W)$, $(-T)(v) = -(T(v))$

$$(T + (-T))(v) = T(v) + (-T)(v) \\ = T(v) - T(v) = 0$$

$$\therefore T + (-T) = 0.$$

(e) $I \in F$

$$(IT)(v) = I(T(v)) = T(v)$$

$$I \cdot T = T$$

$$(f) (a(S+T))(v) = a((S+T)(v)) \\ = a(S(v) + T(v)) \\ = aS(v) + aT(v) \\ = (aS + aT)(v)$$

$$\therefore a(S+T) = aS + aT$$

$$(g) ((a+b)T)(v) = (a+b)T(v) \\ = aT(v) + bT(v) \\ = (aT + bT)(v)$$

$$\therefore (a+b)T = aT + bT$$

$$(h) (a(bT))(v) = a((bT)(v)) = a(bT(v)) \\ = (ab)T(v) = ((ab)T)(v)$$

$$\therefore a(bT) = (ab)T$$

$\therefore L(V, W)$ is a vector space.

Eg 1.17 p.22

Suppose $D \in L(P_3(R), P_2(R))$ is the differentiation map defined by $Dp = p'$. Find the matrix D with respect to the standard bases of $P_3(R)$ and $P_2(R)$.

Soln:

The standard bases of $P_3(R)$ is $1, x, x^2, x^3$.

The standard bases of $P_2(R)$ is $1, x, x^2$.

$$D(1) = 0 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2$$

$$D(x) = 1 = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2$$

$$D(x^2) = 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2$$

$$D(x^3) = 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2$$

$$M(D) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

S.A 1-3 No.11 p.31 check

Suppose $a = (a_1, \dots, a_n)$ is 1-by-n matrix and C is n-by-p matrix. Prove that

$$aC = a_1C_1 + \dots + a_nC_n.$$

Soln:

$$a = (a_1 \ a_2 \ \dots \ a_n)$$

$$C = \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{pmatrix} = \begin{pmatrix} C_{1,1} & C_{1,2} & \dots & C_{1,p} \\ C_{2,1} & C_{2,2} & \dots & C_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n,1} & C_{n,2} & \dots & C_{n,p} \end{pmatrix}$$

$$aC = (a_1 \ a_2 \ \dots \ a_n) \begin{pmatrix} C_{1,1} & C_{1,2} & \dots & C_{1,p} \\ C_{2,1} & C_{2,2} & \dots & C_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n,1} & C_{n,2} & \dots & C_{n,p} \end{pmatrix}$$

$$= ((a_1 C_{1,1} + a_2 C_{2,1} + \dots + a_n C_{n,1}) \ \dots \ (a_1 C_{1,2} + a_2 C_{2,2} + \dots + a_n C_{n,2}) \ \dots \ (a_1 C_{1,p} + a_2 C_{2,p} + \dots + a_n C_{n,p}))$$

$$= (a_1 C_{1,1} \ a_1 C_{1,2} \ \dots \ a_n C_{1,p}) \\ + (a_2 C_{2,1} \ a_2 C_{2,2} \ \dots \ a_2 C_{2,p}) + \dots \\ + (a_n C_{n,1} \ a_n C_{n,2} \ \dots \ a_n C_{n,p})$$

$$= a_1 (C_{1,1} \ C_{1,2} \ \dots \ C_{1,p}) \\ + a_2 (C_{2,1} \ C_{2,2} \ \dots \ C_{2,p}) + \dots \\ + a_n (C_{n,1} \ C_{n,2} \ \dots \ C_{n,p}) \\ = a_1 C_1 + a_2 C_2 + \dots + a_n C_n.$$